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Enhancement of hydrogen evolution activity of CoS_2/MoS_2 heterostructure catalysts by electronic structure modulation

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ABSTRACT

MoS_2 exhibits high catalytic ability; however, its wide band gap (1.73 eV) and poor intrinsic conductivity significantly restrict the enhancement of its catalytic activity. To enhance the electron transfer between the interface of heterojunctions and improve the hydrogen evolution reaction (HER) performance, the paper constructs a CoS_2/MoS_2 heterostructures by growing CoS_2/MoS_2-X on carbon cloth (CC) using a straightforward one-step method. It investigates the HER properties of the heterojunctions of various molar ratios of ammonium molybdate to cobalt nitrate. The results indicate that the CoS_2/MoS_2-X catalyst exhibits excellent electrocatalytic activity (73 mV@10 mA cm^{-2}) and Tafel slope (80.42 mV dec^{-1}) at 1.0 M KOH when the molar ratio of ammonium molybdate to cobalt nitrate is 1:1 (X = 5). CoS_2/MoS_2-5 exhibits good durability of over 100 h when current densities are 10 and 100 mA cm^{-2}. Density functional theory (DFT) calculations indicate that heterostructures present an optimized Gibbs free energy of hydrogen adsorption (ΔG_{H^*}) close to zero, providing insight into the exceptional HER performance. This work provides a simple, effective and innovative strategy for preparing high-activity CoS_2/MoS_2 catalysts by modulation of electronic structure.

1. Introduction

Electrochemical water splitting, as an advanced clean and sustainable hydrogen production technology, has received increasing attention and research from scholars [1,2]. The slow kinetics of the electrode reaction necessitate the application of catalyst technology, and the most effective electrocatalysts now are often materials based on noble metals (Pt/C) [3,4]. However, its high cost and scarce resources have hindered its commercial application in hydrogen evolution reaction (HER) [5]. Therefore, the development of highly active, stable and earth-abundant material substitutes has become a hot research topic, including monatomic catalysts [6,7], metal oxides [8,9], hydroxides [10,11], sulfides [12,13], nitrides [14,15], selenides [10,16], phosphides [17,18], and so on.

Due to its distinctive electronic configuration in three-dimensional orbitals, intriguing structural and performance characteristics, and appropriate H-absorption energy in the HER, molybdenum sulfide (MoS_2) has garnered significant interest [19–21]. Unfortunately, as a typical semiconductor material, the poor intrinsic conductivity of MoS_2 limits the electron transfer in the catalytic reaction, thereby affecting the HER activity [22]. Although the ultrathin thickness and 2D morphology of MoS_2 layered structures allow for a large surface area, the active sites for HER are predominantly located at the edges of the layers rather than at the basal surface [19]. Poor electrical conductivity [23] and inadequate exposure to the electroactive sites [24,25] result in MoS_2 not functioning effectively as a HER catalyst.

Interface engineering is an effective strategy to enhance the activity of MoS_2 catalysts in alkaline media. One research focus is the construction of MoS_2-based heterostructure catalysts, which promote the initial hydrolysis dissociation process by providing additional active sites to accelerate reaction kinetics and improve reaction stability through the complementary advantages and synergistic effects at the interfaces of various phase catalysts within the heterostructures [26]. For instance, Zhang et al. designed and synthesized a WC_{1-x}/Mo_2C@CNF heterostructure catalyst, in which the formation of a built-in electric field (BEF) drives the migration of electrons from Mo_2C to WC_{1-x}. This process leads to an asymmetric charge distribution and a shift in the d-band center, effectively optimizing intermediates'

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adsorption/desorption strength [27]. Yang et al. synthesized $CoTe_2$/CoP composite materials with heterointerfaces, where the built-in electric field drives the transfer of electrons from $CoTe_2$ to CoP at the interface, increasing the electron density on the Co atoms in CoP, achieving electron redistribution and optimizing the adsorption free energy of reaction intermediates, thereby accelerating the kinetics of CoTe [28]. Wang et al. synthesized Co-WS_2 nanofiber catalysts, where the work function at the catalyst interface induced a built-in electric field that facilitates the transfer of electrons from Co to WS_2, significantly reducing the energy required for water dissociation and optimizing the Gibbs free energy for the entire reaction process [29]. Doping MoS_2 with transition metals is a potent strategy for enhancing catalytic activity, achieved by fine-tuning electron distribution for improved reaction kinetics [30]. For example, Shi et al. [31] found that Zn-doped MoS_2 (Zn-MoS_2) exhibited excellent electrochemical activity towards HER. The significant enhancement can be attributed to the synergistic effect of the electronic effect (energy level matching) and morphological effect (rich active sites), leading to thermodynamic and kinetic acceleration, respectively. Single Co atom-doped MoS_2 as Deng et al. [32] verified, can provide moderate free energy for H* (ΔG_{H^*}) adsorption and demonstrate good stability in the HER. Co constructs heterostructures to modulate the electronic properties of MoS_2 and enhance HER performance [32,33]. CoS_2 exhibits the highest electrical conductivity among cobalt sulfide (Co_xS_y) compounds, demonstrating superior electrical conductivity and thermal stability [34,35]. Therefore, constructing a heterostructure catalyst of CoS_2 and MoS_2 is a feasible method to enhance the electrocatalytic activity of the MoS_2 material.

In this paper, a series of heterostructure nanosheet electrocatalysts (CoS_2/MoS_2-X) with controllable composition and morphology were prepared for the HER using a simple one-step hydrothermal method by adjusting the molar ratio of ammonium molybdate to cobalt nitrate. The CoS_2/MoS_2-5 demonstrated a strong chemisorption of hydrogen intermediates during hydrolysis dissociation. This enhancement significantly improved the HER electrocatalytic activity (73, 112, and 144 mV@10, 20, and 50 mA cm^{-2}) and Tafel slopes (80.42 mV dec^{-1}) in 1.0 M KOH. CoS_2/MoS_2-5 provided a larger surface area, more active sites, and smaller charge transfer resistance, substantially improving activity towards the HER. In addition, CoS_2/MoS_2-5 demonstrated good durability, lasting over 100 h when current densities are 10 and 100 mA cm^{-2}. The hydrogen adsorption Gibbs free energies of CoS_2, MoS_2, and CoS_2/MoS_2 were analyzed by density functional theory (DFT) calculations, and the underlying reasons for the excellent performance were illustrated. The prepared CoS_2/MoS_2-5 heterojunction nanosheets demonstrated excellent performance as HER catalysts, suggesting that the low-cost heterostructure compounds have a promising application in the alkaline water electrolysis of hydrogen.

2. Experimental section

2.1. Chemicals and reagents

Thiourea ($CS(NH_2)_2$), ammonium molybdate ($(NH_4)_6Mo_7O_{24}\bullet 4H_2O$), cobalt nitrate hexahydrate ($Co(NO_3)_2{\cdot}6H_2O$), potassium hydroxide (KOH), anhydrous ethanol (CH_3CH_2OH), acetone (C_3H_6O), and concentrated nitric acid (HNO_3) were all of analytical grade and used directly. Carbon cloth was purchased from Suzhou Zhengtai Rong New Material Co., Ltd. The water used in all experiments was deionized water. Before use, the carbon cloths were sequentially sonicated with acetone, anhydrous ethanol, and deionized water for 15 min and then soaked in concentrated nitric acid for 12 h. The carbon cloths, which were of analytical grade, were used directly.

2.2. Synthesis of CoS_2/MoS_2

CoS_2/MoS_2 were grown on carbon cloth by a one-step hydrothermal method. First, 30 mmol of thiourea ($CS(NH_2)_2$), x mmol of cobalt nitrate hexahydrate ($Co(NO_3)_2\bullet 6H_2O$), and 2−x mmol of ammonium molybdate ($(NH_4)_6Mo_7O_{24}\bullet 4H_2O$) were added sequentially in 25 ml of deionized water. The solution was sonicated and stirred until it became homogeneous. Then the solution was transferred to a 50 ml PTFE-lined stainless steel autoclave containing a clean carbon cloth and heated at 150 °C for 24 h. After that, allow the carbon cloth to naturally cool to room temperature at the end of the reaction, then remove the carbon cloth and rinse it alternately with deionized water and ethanol. Finally, the carbon cloth was dried in a vacuum oven at 60 °C for 6 h. A series of samples was obtained by adjusting the molar ratios (x) of ammonium molybdate ($(NH_4)_6Mo_7O_{24}\bullet 4H_2O$) and cobalt nitrate hexahydrate ($Co(NO_3)_2\bullet 6H_2O$) in solution to be 10:0, 9:1, 8:2, 7:3, 6:4, 5:5, 4:6, 3:7, 2:8, 1:9, and 0:10 (noted as MoS_2, CoS_2/ MoS_2-1, CoS_2/MoS_2-2, CoS_2/MoS_2-3, CoS_2/MoS_2-4, CoS_2/MoS_2-5, CoS_2/MoS_2-6, CoS_2/MoS_2-7, CoS_2/MoS_2-8, CoS_2/MoS_2-9, CoS_2, respectively).

2.3. Material characterization

The X-ray diffraction patterns (XRD) of the resultant samples are detected from a Smart Lab X-ray diffractometer which uses Cu Kα radiation as an X-ray source (Rigaku Corporation, Japan, λ = 1.5418 Å), and the measurement range of 2θ is 10–80°. The microstructure analysis is explored by Merlin Compact scanning electron microscope (Carl Zeiss NTS GmbH, Germany). The transmission electron microscopy (TEM) images and elementary composition of the experimental samples are detected using a JEOL JEM 2100F (Japan) high-resolution transmission electron microscope which is armed with energy dispersive spectroscopy (EDS). The Thermo ESCALAB 250XI (Thermo Scientific K-Alpha, America) X-ray photoelectron spectroscopy (XPS) with Al Kα (1486.6 eV) X-ray source is carried out to ascertain the chemical states of elements on the surface of the sample. The inductively coupled plasma optical emission spectrometer (ICP-OES) measurements were conducted using an Agilent 5110 from the United States. The alpha300R of WITec (Germany) is used to perform Raman spectroscopy tests on samples.

2.4. Electrochemical measurements

All electrochemical measurements were performed in a three-electrode system using an electrochemical workstation (CHI760E) with the prepared self-supporting electrodes directly as the working electrode, a graphite rod as the auxiliary electrode, and a saturated calomel electrode (SCE) as the reference electrode in N_2 saturated 1.0 M KOH. The geometric surface area of the working electrode is 1 cm × 1 cm. According to the previous study, the SCE was calibrated concerning RHE: in 1.0 M KOH, $E_{RHE} = E_{SCE} + 0.059 \times pH + 0.242$. The HER test was performed in N_2-saturated 1.0 M KOH aqueous solution. LSV polarization and Tafel plots were obtained at a scan rate of 2 mV s^{-1} with 95 % IR compensation. The impedance of the catalytic electrode during the reaction and the reaction kinetics was measured by AC impedance method. The high and low-frequency range is 10^5–0.01 Hz, and the applied bias voltage is − 1.26 V (vs. SCE) in the HER responses. Electrochemical Active Surface Area (ECSA) was measured in the non-Faraday potential region at different scan rates (1, 2, 4, 6, 8 and 10 mV s^{-1}). The HER stability of the working electrode was tested by chronopotentiometry at current densities of 10 and 100 mA cm^{-2}.

2.5. Density functional theory (DFT) calculations

DFT calculations were performed in the Vienna ab initio simulation package (VASP) [36], and the results were post-processed using the Vaspkit [37] program package. In the structure optimization and self-consistent calculations, the electronic interactions are described using the GGA-PBE (Perdew-Burke-Ernzerhof of generalized gradient exchange approximation) [38] exchange-correlation generalization with the spin polarization enabled. The kinetic energy truncation energy is set to 520 eV for all plane-wave basis sets. The Brillouin zone was

delineated using 6 × 6 × 6, 8 × 8 × 1 and 4 × 4 × 1 Monkhorst-Pack grids for the primitive cells and heterojunctions, respectively. The optimization was performed using the CG (Conjugate gradient) algorithm with Van der Waals corrections by the DFT-D3 method, and the convergence of the energy and force parameters were set to 1×10^{-5} and 0.02 eV A^{-1}.

To better understand the catalytic activity of the HER, the hydrogen adsorption energy (ΔE_H) and the free energy of hydrogen adsorption (ΔG_{H^*}) at room temperature (298 K) are defined as:

$$\Delta E_H = E_{total} - E_{slab} - \frac{E_{H_2}}{2} \quad (1)$$

$$\Delta G_{H_*} = \Delta E_H + \Delta E_{zpe} - T\Delta S_H \quad (2)$$

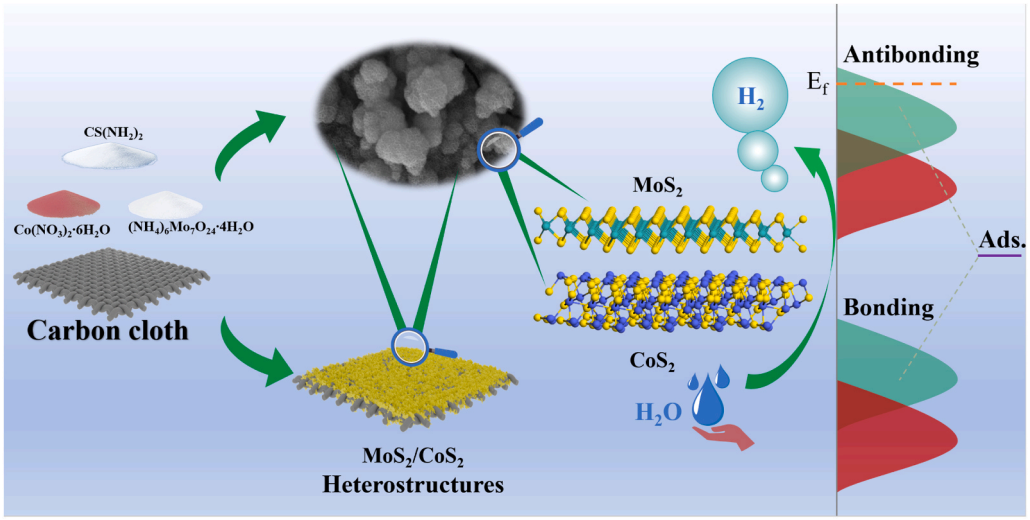
where ΔE_H is the hydrogen adsorption energy, E_{total} is the total energy for the adsorption state, E_{slab} is the energy of the pure surface, E_{H_2} is the energy of H_2 in the gas phase, ΔG_{H_*} is the free energy of hydrogen adsorption, ΔE_{zpe} is the zero-point energy change and ΔS_H is the entropy change.

3. Results and discussion

3.1. Catalyst synthesis and characterization

Heterojunction nanosheet cluster CoS_2/MoS_2-X were prepared by varying the ratio of ammonium molybdate ($(NH_4)_6Mo_7O_{24}\bullet 4H_2O$) and cobalt nitrate hexahydrate ($Co(NO_3)_2{\cdot}6H_2O$) (Scheme 1).

The crystal structure and morphology of CoS_2/MoS_2-5 were studied by X-ray diffraction (XRD), scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS) and transmission electron microscopy (TEM) (Fig. 1). XRD was used to characterize the structural and compositional information of CoS_2/MoS_2-5. The complete diffraction peaks corresponding to the two substances CoS_2 (PDF#65-3322) and MoS_2 (PDF#75-1539) (Fig. 1a), indicating the characteristics of the CoS_2/MoS_2-5 mixed-phase heterojunction. All samples were simultaneously subjected to XRD testing, and it can be observed that the diffraction peaks of the CoS_2 sample perfectly match the standard card PDF#65-3322 of CoS_2, confirming the successful synthesis of CoS_2. The XRD patterns of the CoS_2/MoS_2-9 and CoS_2/MoS_2-8 samples indicate that with the addition of Mo, some of the CoS_2 diffraction peaks are transformed into the diffraction peaks of $CoS_{1.097}$, proving the formation of a small amount of $CoS_{1.097}$ material. Due to the amorphous structure of MoS_2, the diffraction peaks of MoS_2 and CoS_2 in samples such as CoS_2/MoS_2-7, CoS_2/MoS_2-6, CoS_2/MoS_2-5, CoS_2/MoS_2-4, CoS_2/MoS_2-3, CoS_2/MoS_2-2, CoS_2/MoS_2-1, and MoS_2 are not clearly visible, which also leads to the poor display of CoS_2 diffraction peaks (Fig. S1). Therefore, Raman spectroscopy tests were conducted on MoS2 and CoS2/MoS2-5 samples. The MoS_2 sample exhibits characteristic peaks of MoS_2 at 373.22 and 403.44 cm^{-1}, and the CoS_2/MoS_2-5 sample also shows similar characteristic peaks at 373.58 and 404.88 cm^{-1}, indicating the presence of MoS_2 in both MoS_2 and CoS_2/MoS_2-5 samples (Fig. S2). Compared to MoS_2, the characteristic peaks of CoS_2/MoS_2 shift towards a higher wavenumber, suggesting that there is electronic interaction between the components of the heterojunction. The morphology of CoS_2/MoS_2-5 was observed using SEM at different magnifications. The heterojunction material encapsulates the entire carbon fiber and has a three-dimensional structure with a larger surface area. This structure aids in exposing more activity sites and accelerating the penetration of the electrolyte (Fig. 1b), and the heterojunction material is a homogeneous cluster of nanosheets of similar size (Fig. 1c). In addition, SEM observations at different magnifications were conducted on materials such as MoS_2, CoS_2/MoS_2-X (where X is 1, 2, 4, 5, 6, 8, and 9), and CoS_2. Figs. S3 and S4 show that the microstructure of the MoS_2 material is scattered nanosheet-like material. During the preparation process of the materials, as $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$ decreased and $Co(NO_3)_2{\cdot}6H_2O$ increased, a small amount of agglomeration of the scattered nanosheet-like material was observed. When the ratio of $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$ to $Co(NO_3)_2{\cdot}6H_2O$ was 1:1, the agglomeration of nanosheets in the CoS_2/MoS_2-5 material reached its peak. Subsequently, as the amount of $Co(NO_3)_2{\cdot}6H_2O$ added exceeded that of $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$, the agglomeration of nanosheets in the CoS_2/MoS_2-6 material began to decrease. The nanosheets in the CoS_2/MoS_2-8 material had disappeared, and a small number of nanospheres began to appear. The morphology of the CoS_2/MoS_2-9 material changed from a small number of nanospheres to a large number of irregular nanoblocks, and the size of the nanoblocks was smaller than that of the nanospheres. Finally, when the preparation of the material had no $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$ and only $Co(NO_3)_2{\cdot}6H_2O$, the morphology of the CoS_2 material was blocky objects that were stuck together and did not fully cover the carbon fibers. Through the analysis of scanning electron microscopy at different magnifications of MoS_2, CoS_2/MoS_2-X (where X is 1, 2, 4, 5, 6, 8, and 9), and CoS_2 materials, it was concluded that when the ratio of $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$ to $Co(NO_3)_2{\cdot}6H_2O$ was 1:1, the CoS_2/MoS_2-5 material had well-developed sheet-like structures due to nanosheet agglomeration, which may have the best hydrogen evolution activity. TEM images show a hybrid structure of flower-like CoS_2/MoS_2-5 heterostructures with nanosheet crossing edges (Fig. 1d). HRTEM images show different interfaces on CoS_2/MoS_2-5 heterostructured nanosheets, and the lattice fringes with



Scheme 1. Schematic representation of the preparation of CoS_2/MoS_2 heterostructure materials.

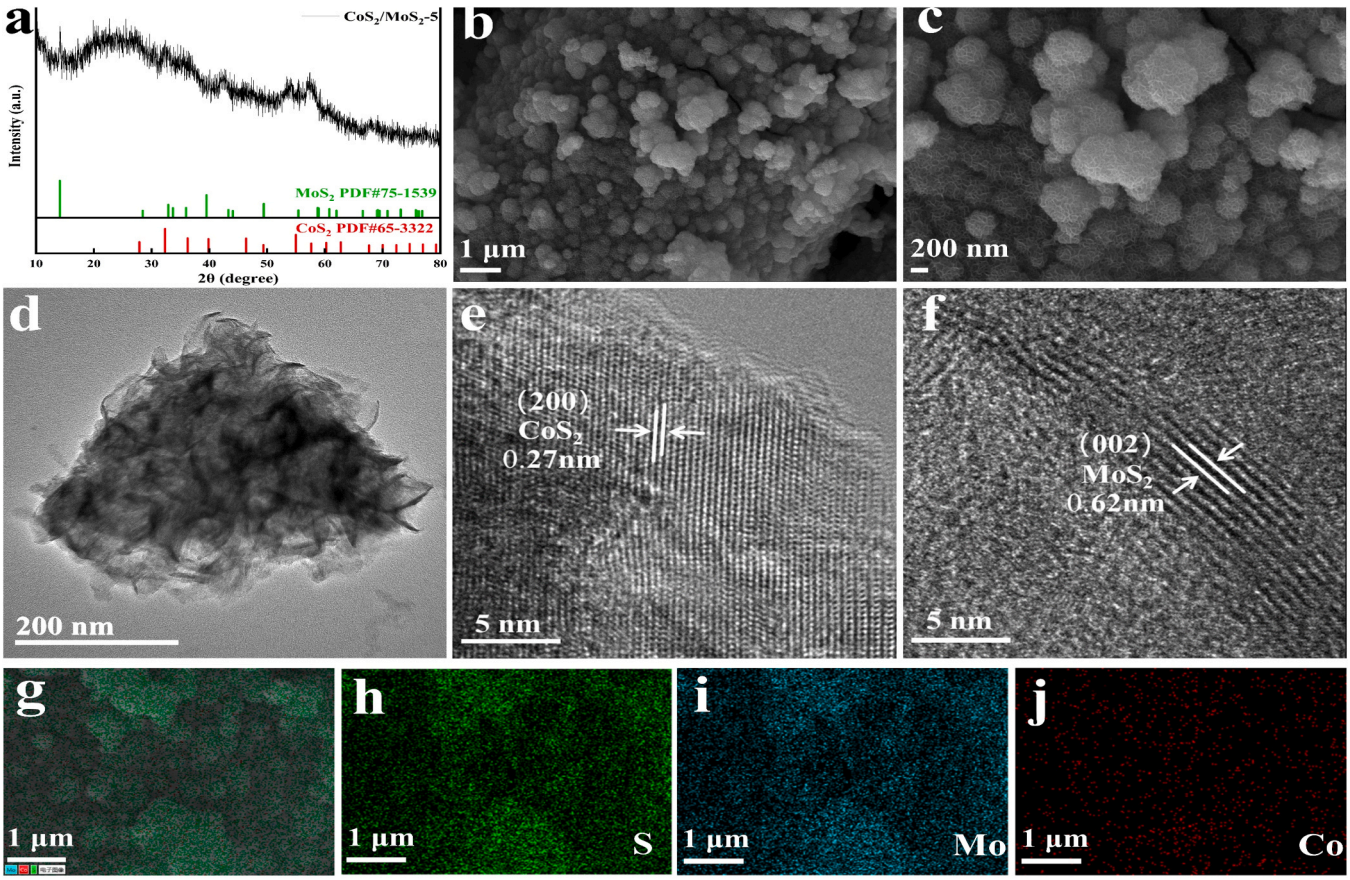


Fig. 1. (a) XRD pattern of CoS_2/MoS_2-5 catalyst; (b–c) SEM images of CoS_2/MoS_2-5 catalyst; (d) TEM images of CoS_2/MoS_2-5 catalyst; (e–f) HRTEM image of CoS_2/MoS_2-5 catalyst; (g–j) EDS mapping obtained by SEM for the selected area of CoS_2/MoS_2-5 catalyst.

interplanar spacings of 0.27 and 0.62 nm correspond to the (200) crystal plane of CoS_2 and the (002) crystal plane of MoS_2, respectively (Figs. 1e and 1f). The results of the crystal planes obtained by TEM and XRD are consistent, which proves the CoS_2/MoS_2-5 are heterostructured nanosheets. The EDS elemental mapping images clearly reveal that Mo (blue), Co (red), and S (green) are uniformly distributed on the surface of the lamellar nanostructures (Fig. 1g–j), indicating the homogeneous distribution of these elements in CoS_2/MoS_2-5. The CoS_2 and MoS_2 phases are tightly bonded and homogeneously distributed, with no obvious phase separation. The molar ratio of MoS_2 and CoS_2 is about 3:1 based on the ICP-OES in Table S1. According to the aforementioned analyses, the heterostructure $CoS_2/MoS2–5$ was successfully prepared.

The elemental composition and valence states of the CoS_2/MoS_2-5 heterojunction electrocatalyst were further investigated by XPS (Fig. 2), revealing the presence of Co, Mo, and S elements (Fig. 2a) without any other impurity interferences. The high-resolution Co 2p XPS spectrum (Fig. 2b) shows two main peaks at binding energies of 782.65 and 798.44 eV, corresponding to the $2p_{3/2}$ and $2p_{1/2}$ peaks of Co^{2+} in CoS_2/MoS_2-5, and two additional peaks at 786.52 and 802.31 eV, attributed to satellite peaks of Co^{2+}. The peaks in the Co 2p spectrum near 779.38 and binding energy near 794.37 eV can be attributed to $2p_{3/2}$ and $2p_{1/2}$ of Co^{3+} [39–41]. It is noteworthy that the Co^{2+} $2p_{3/2}$ peak in CoS_2/MoS_2-5 exhibits a notable shift towards higher binding energy by 1.77 eV compared to pure CoS_2, accompanied by a significant enhancement in the peak intensity of the Co^{2+} species. This observation underscores the reduction in the valence state of Co species upon integration with MoS_2, suggesting a potential shift from higher valence states towards the more stable + 2 state.

The Mo 3d XPS spectrum of the CoS_2/MoS_2-5 catalyst (Fig. 2c) shows that the peaks with binding energies of 228.99 and 232.14 eV correspond to Mo^{4+} $3d_{5/2}$ and Mo^{4+} $3d_{3/2}$ in CoS_2/MoS_2-5, respectively. The Mo^{6+} signal at 233.43 eV may have been generated by the oxidation of Mo^{4+} exposed to air. Compared with pure MoS_2, the binding energy of Mo^{4+} $3d_{5/2}$ in CoS_2/MoS_2-5 shifts negatively by 0.21 eV, indicating a strong electronic coupling between CoS_2 and MoS_2 in the CoS_2/MoS_2-5 heterojunction. Furthermore, the S 2s signal of the CoS_2/MoS_2-5 at a binding energy of 226.05 eV, adjacent to the Mo 3d region, indicates the presence of Co-S and Mo-S bonds, which induce strong electronic interactions at the heterojunction interface.[39]

The S 2p spectra of the CoS_2/MoS_2-5 catalyst (Fig. 2d) can be deconvoluted into four peaks. The peaks centered at binding energies of 161.75 and 162.97 eV correspond to the S^{2-} $2p_{3/2}$ and S^{2-} $2p_{1/2}$ of the MoS_2 in CoS_2/MoS_2-5, respectively. The peak located at 163.53 eV corresponds to the S_2^{2-} of CoS_2 in the CoS_2/MoS_2-5. The peak at 169.32 eV originates from the surface oxidation product SO_4^{2-}. Compared with pure CoS_2 and MoS_2, the S 2p peak in the CoS_2/MoS_2-5 heterostructure shifts towards a lower binding energy, indicating mutual electron transfer between MoS_2 and CoS_2. The above results suggest that there is a strong electronic interaction at the CoS_2/MoS_2-5 heterostructure interface, and the restructured electronic structure through interface engineering enhances the electrocatalytic activity of the CoS_2/MoS_2-5 heterostructure for hydrogen evolution during water electrolysis.

3.2. Hydrogen evolution reaction measurement

The HER activity of a series of samples CoS_2/MoS_2-X was initially evaluated using a three-electrode configuration in 1.0 M nitrogen-saturated KOH. A slow scan rate (2 mV s^{-1}) was employed to minimize the background current, and all potentials are referenced to the

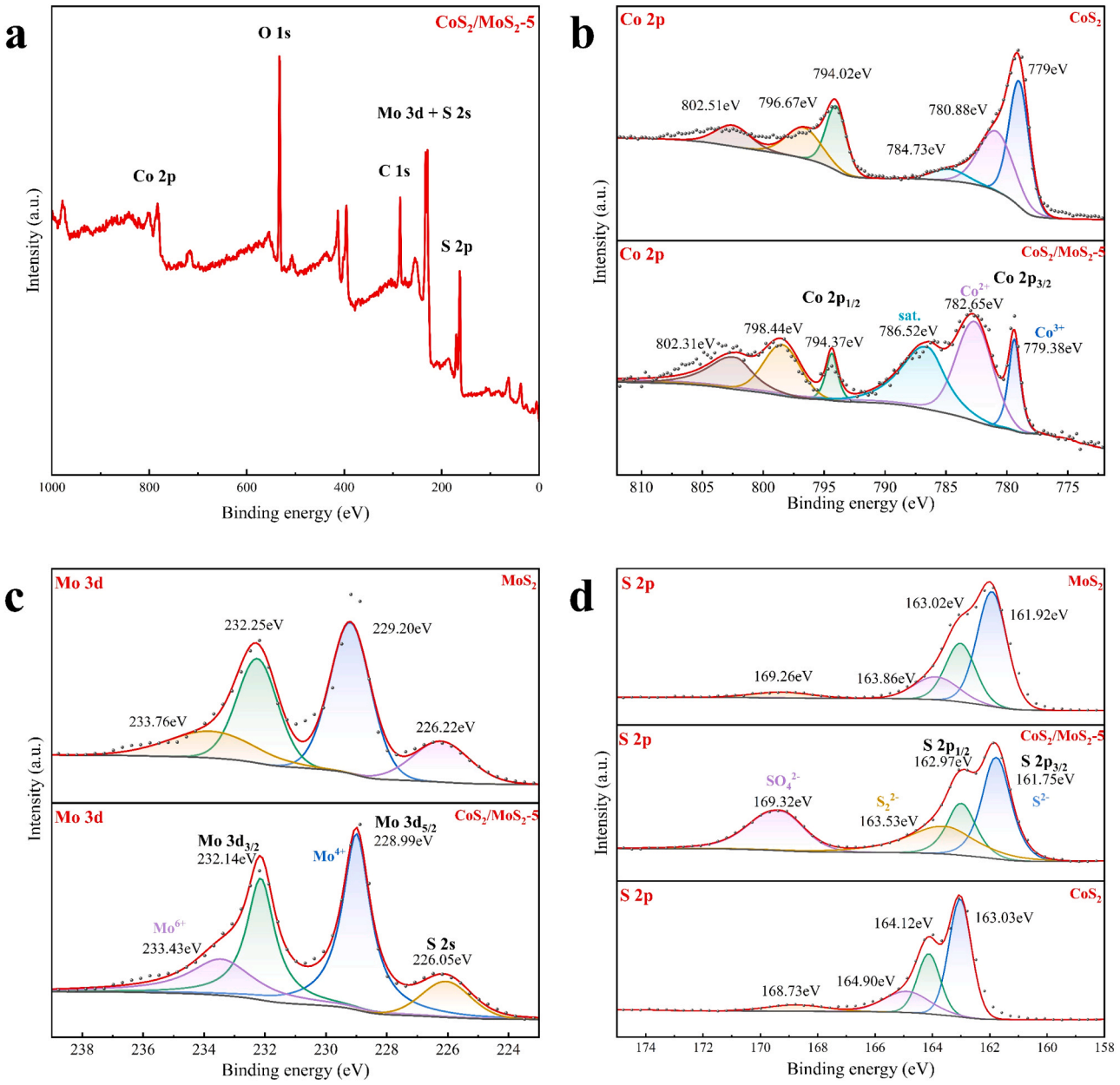


Fig. 2. (a) Full XPS spectrum of CoS_2/MoS_2-5 catalyst; (b) High-resolution XPS spectra of Co 2p; (c) High-resolution XPS spectra of Mo 3d; (d) High-resolution XPS spectra of S 2p.

standard reversible hydrogen electrode (RHE). The effects of different molar ratios of $(NH_4)_6Mo_7O_{24}•4H_2O$ to $Co(NO_3)_2·6H_2O$ on the HER performance of the prepared catalysts were investigated when the total is 2 mmol, including 10:0, 9:1, 8:2, 7:3, 6:4, 5:5, 4:6, 3:7, 2:8, 1:9, and 0:10. The IR-compensated polarization curves of the four samples (Pt/C, MoS_2, CoS_2, and CoS_2/MoS_2-5) show that the CoS_2/MoS_2-5 catalyst exhibits excellent catalytic performance, second only to Pt/C (Fig. 3a).

Moreover, the hydrogen evolution performance of the CoS_2/MoS_2-5 catalyst is the best among the series of CoS_2/MoS_2-X catalysts prepared (Fig. S5a). Based on the column plots of overpotentials at different current densities for the aforementioned four samples (Fig. 3b), the CoS_2/MoS_2-5 catalyst demonstrated superior HER electrocatalytic activity compared to Pt/C, with overpotentials of 73 mV at 10 mA cm^{-2}, 112 mV at 20 mA cm^{-2}, and 144 mV at 50 mA cm^{-2}, respectively. The CoS_2/MoS_2-X catalysts also showed the smallest overpotentials in the column comparison plots of the series (Fig. S5b).

In addition, when only ammonium molybdate or cobalt nitrate was added during the preparation of the catalysts, the HER electrocatalytic performances of MoS_2 and CoS_2 were at their worst. However, with the change in the molar ratio of ammonium molybdate to cobalt nitrate (Fig. 3c), the HER electrocatalytic performances of the prepared catalysts changed (Fig. S5c). The HER reaction kinetics at the catalytic electrode can be observed through the Tafel slope derived from the IR-compensated polarization curve. The Tafel slope of the CoS_2/MoS_2-5 electrode is 80.42 mV dec^{-1} (Fig. 3d), which is the smallest among all the catalysts except for Pt/C. From the Tafel slopes of CoS_2/MoS_2-X catalysts and the bar plots (Figs. S5c and S5d), the CoS_2/MoS_2-5 electrode exhibits the highest reaction rate and the best reaction kinetics in the HER process. Moreover, the Tafel slopes are consistent with the overpotential results obtained by LSV, demonstrating that CoS_2/MoS_2-5 exhibits excellent catalytic performance.

To further investigate the HER activity of CoS_2/MoS_2-X samples, ECSA and electrochemical impedance spectroscopy (EIS) were measured (Fig. 4). The electrochemical active area was evaluated by the

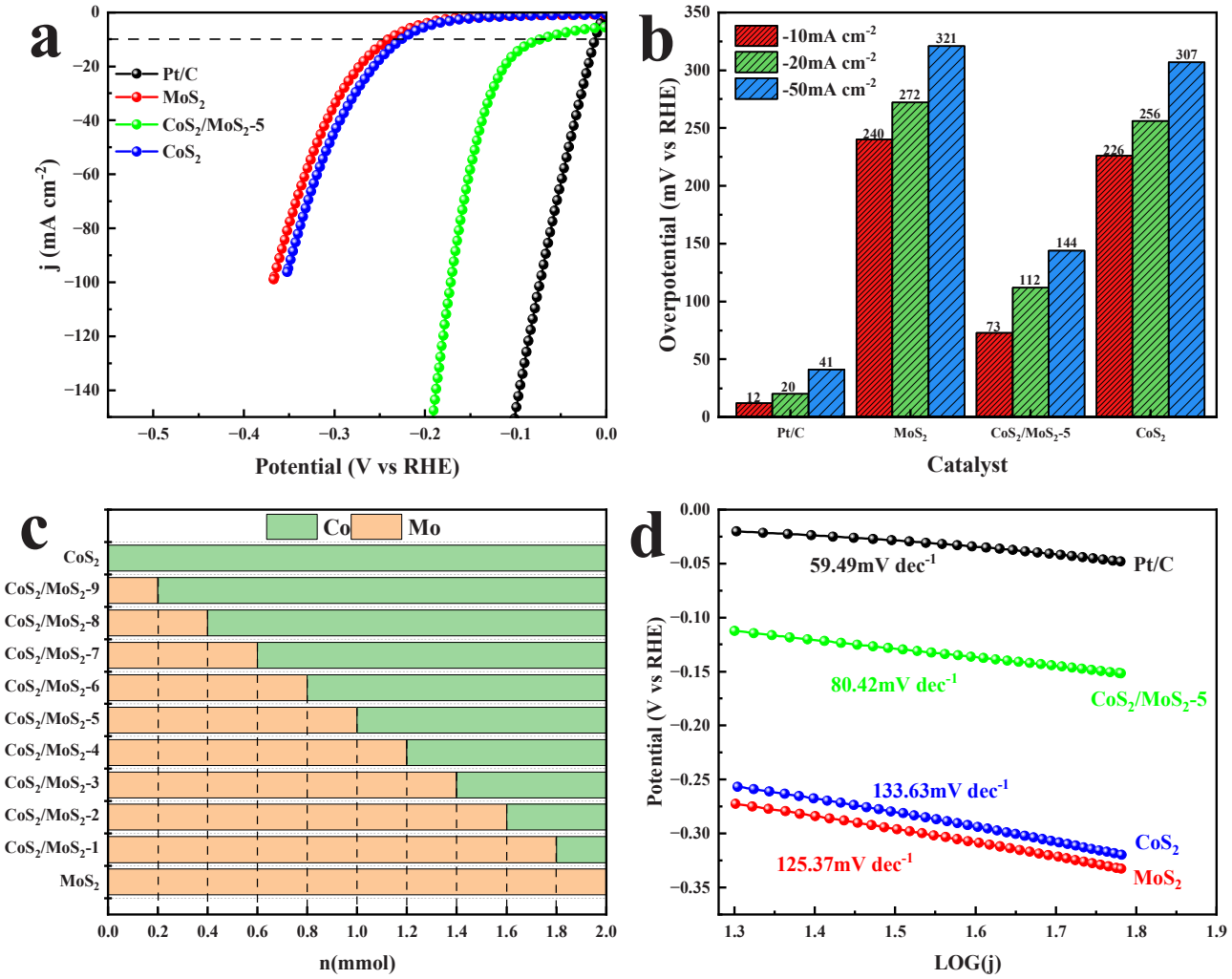


Fig. 3. (a) Linear sweep voltammetry curves of CoS_2/MoS_2-5 and other catalysts; (b) Column comparison of overpotentials at different current densities of CoS_2/MoS_2-5 and other catalysts; (c) Molar ratios of a series of catalysts of CoS_2/MoS_2-X; (d) Comparison of Tafel slopes of CoS_2/MoS_2-5 and other catalysts.

C_{dl} values obtained from cyclic voltammetry (CV) curves at various scan rates.

Based on the electrochemical impedance spectra of the four samples, Pt/C, MoS_2, CoS_2 and CoS_2/MoS_2-5 (Fig. 4a), and the magnified details (Fig. 4b), we can determine the charge transfer capacity of the catalyst. The high-frequency intersection of the curves with the x-axis, as shown in Figs. 4a and 4b represent the solution resistance (Rs), mainly caused by the electrolyte and the exposed catalyst substance. The diameter of the semicircle represents the charge transfer resistance (R_{ct}). To separate the ohmic resistance from the polarization resistance, we use an equivalent circuit model in which R_s is connected in series with a module, and R_{ct} is connected in parallel with the constant phase element (CPE) (Fig. 4a). The R_{ct} values for a series of MoS_2, CoS_2/MoS_2-X (1, 2, …, 8, 9), and CoS_2 catalysts are detailed in Figs. S6a and S6b, with values of 26.14, 4.616, 2.387, 1.704, 1.629, 1.199, 1.48, 1.764, 2.244, 15.52, and 18.58 Ω, respectively. This indicates that CoS_2/MoS_2-5 has the smallest charge transfer resistance at 1.199 Ω, suggesting excellent electrical conductivity and electrochemical properties for alkaline hydrogen evolution. Consequently, this can significantly improve the charge transfer rate in electrolytic water for hydrogen evolution.

The capacitance of the double electric layer (C_{dl}) from four samples of Pt/C, MoS_2, CoS_2, and CoS_2/MoS_2-5 can be used to evaluate the specific surface area of the catalysts (Fig. 4c). The C_{dl} values (Figs. S6c and S6d) were obtained by linearly fitting the CV curves (Fig. S7 a–l) for a series of CoS_2/MoS_2-X catalysts, it was observed that the C_{dl} value of CoS_2/MoS_2-5 is the highest among all catalysts (Fig. 4d), at 456.84 mF cm^{-2}, indicating that the heterogeneous structure of CoS_2/MoS_2-5 can expose more active sites, leading to the best hydrogen production performance. The above results indicate that the best performance of CoS_2/MoS_2-5 catalysts was obtained when the ratio of ammonium molybdate to cobalt nitrate was 1:1.

Notably, the stability of the CoS_2/MoS_2-X catalysts was also tested (Figs. 5a and S8). During the initial 3600 s, the potential of the CoS_2/MoS_2-X catalysts was found to be relatively stable. The CoS_2/MoS_2-5 catalyst exhibited the most effective hydrogen evolution performance, second only to Pt/C. This observation is consistent with the hydrogen evolution performance indicated by the polarization curves, suggesting that the CoS_2/MoS_2-5 catalyst demonstrates commendable short-term stability.

Most importantly, during hydrogen evolution at stepped current densities ranging from 10 to 200 mA cm^{-2}, the potential changes were found to be stable (Fig. 5b), indicating that the CoS_2/MoS_2-5 catalysts exhibit excellent stability at both low and high current densities. Furthermore, the CoS_2/MoS_2-5 catalyst underwent a 1000-cycle cyclic voltammetry stability test, demonstrating excellent electrocatalytic activity after 1000 CV cycles (Fig. S9). The CoS2/MoS2-5 catalyst morphology after the stability test was also observed. This observation reveals that after the repeatability test, the nanosheet morphology of the material became blunted, resulting in a reduction of its surface area. Nonetheless, when coupled with the findings from the stability test, it

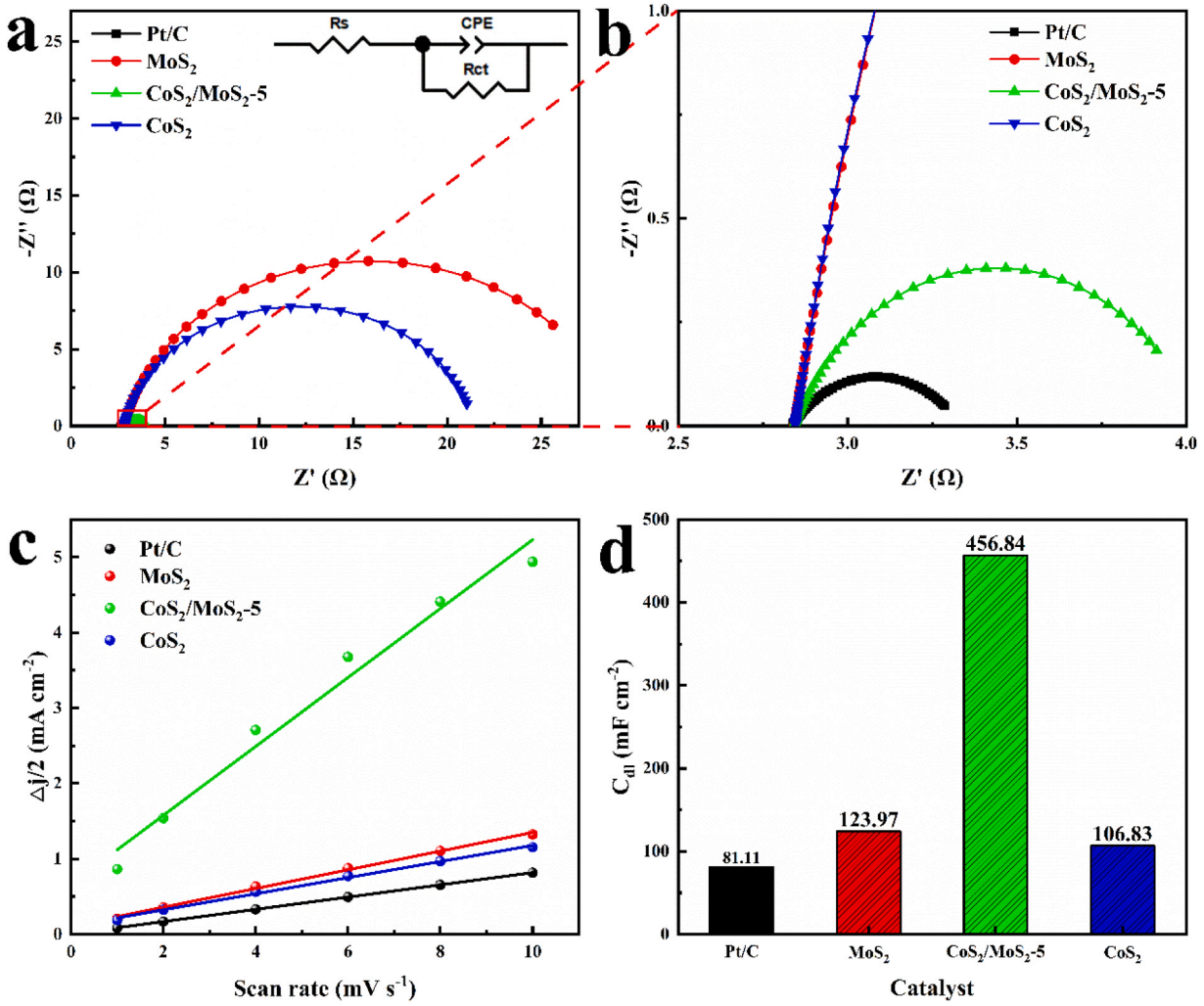


Fig. 4. (a) Electrochemical impedance plots; (b) Enlarged view of electrochemical impedance details; (c) C_{dl} plots; (d) Colum plots of C_{dl} values.

can be concluded that the impact on the catalytic activity of CoS_2/MoS2–5 remains minimal (Fig. S10).

To assess its long-term stability, the CoS2/MoS2-5 catalyst was finally tested for HER stability over 100 h by chronoamperometry at constant current densities of 10 and 100 mA cm^{-2} (Fig. 5c). The overpotential of CoS_2/MoS_2-5 remained essentially unchanged during the test. The excellent performance and stability of CoS_2/MoS2–5 self-supported catalysts are crucial for their practical application in alkaline electrolytic hydrogen production.

Table 1 shows the CoS_2/MoS_2 heterostructured catalysts prepared by researchers in recent years. By comprehensively comparing the parameters such as the complexity of the preparation method, the overpotential at a current density of 10 mA cm^{-2}, the Tafel slope, the charge transfer resistance, and the C_{dl} value, it can be concluded that the CoS_2/MoS_2-5 heterostructured catalyst prepared in this paper exhibits excellent hydrogen evolution performance.

3.3. First-principles calculations

To deeply investigate the enhancement of electrochemical performance by CoS_2/MoS_2 heterojunctions, the energy band structure, density of states, and Gibbs free energy of hydrogen adsorption for the HER of MoS_2, CoS_2, and CoS_2/MoS_2 heterojunctions were calculated by DFT. Fig. 6 displays the molecular models of MoS_2 crystals, CoS_2 crystals, and CoS_2/MoS_2 heterojunctions after structural optimization. MoS_2 crystals have a graphene-like structure, with two layers of S atoms sandwiching a layer of Mo atoms to form a hexagonal filler layer structure, which belongs to the hexagonal crystal system. After orienting them, we selected the (002) face of MoS_2 as the active surface to investigate the catalytic properties of the HER as well as the substrate of the heterojunction. CoS_2 crystals belong to the cubic crystal system, and we selected the (200) face of CoS_2 as the active surface for studying the HER catalytic properties. Meanwhile, the active sites for water dissociation and hydrogen adsorption were calculated. It was found that water molecules would first need to adsorb on the catalyst's surface. The calculation results showed that the adsorption site of water molecules was the Co site on CoS_2 (Fig. S11a), with an adsorption energy of − 0.92 eV, which is the active site for water dissociation. There are three possible adsorption sites for active hydrogen, as shown in Fig. S11b–d: the S-top site on MoS_2, the Co-bri site on CoS_2, and the Co-tri site at the interface.

Fig. 7a–c illustrate the energy band structures of MoS_2, CoS_2, and CoS_2/MoS_2 heterojunctions. MoS_2, with a lamellar crystal structure, is currently the most frequently investigated HER catalyst [49]. Unfortunately, its HER catalytic activities are not excellent due to its poor conductivity. MoS_2 has a forbidden bandwidth of 1.42 eV, and its highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) are at the same K-point, which is typical of direct bandgap semiconductors. This finding agrees with the research conducted by Mak et al. [50] in 2010. The construction of heterostructures has been proven to be an effective pathway to boost the HER performance of MoS2 further, enhancing electron transfer between each component [51].

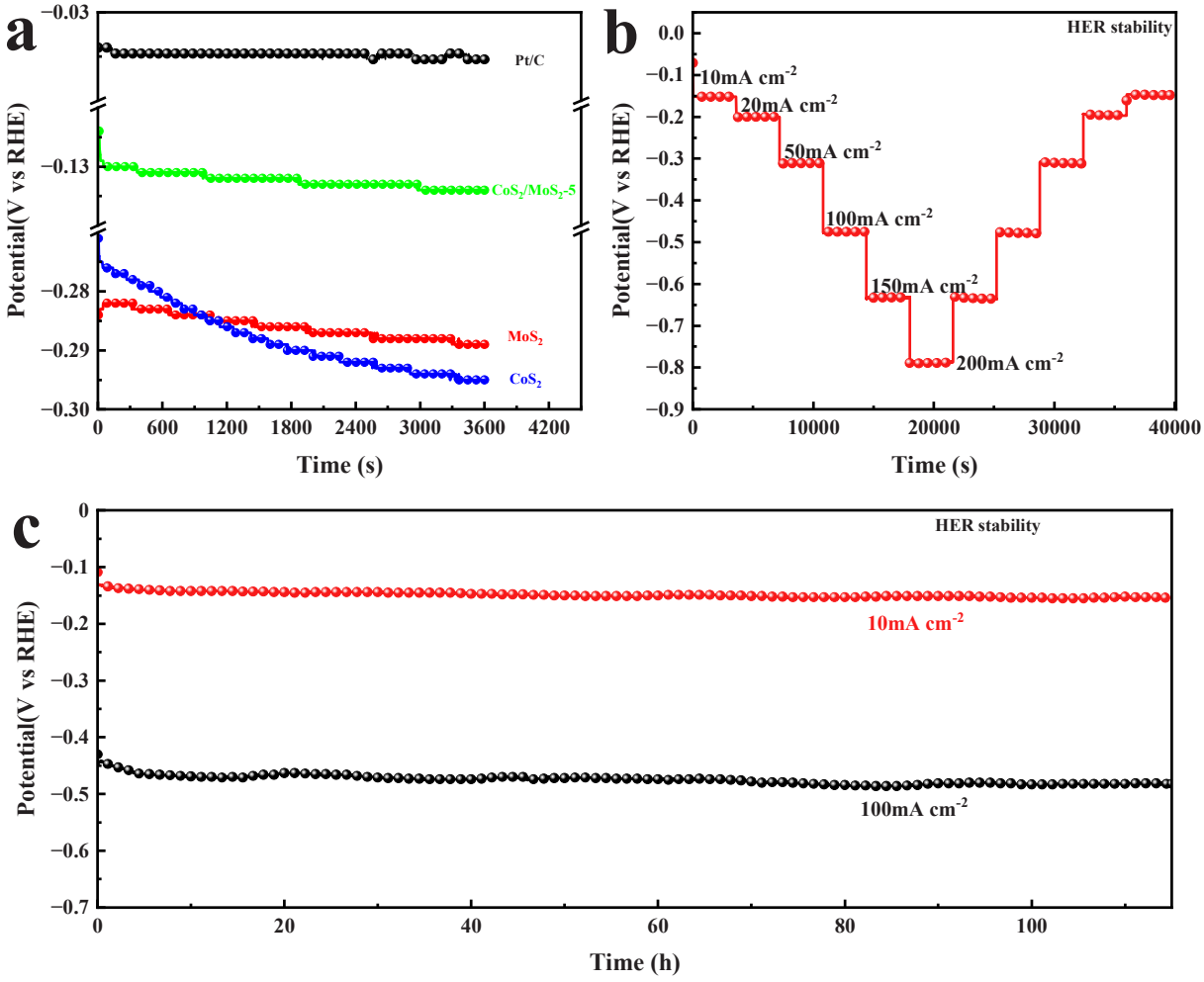


Fig. 5. (a) Short-term stability test of catalysts; (b) Step current operation test of CoS_2/MoS_2-5 catalysts; (c) Long-term stability test of CoS_2/MoS_2-5 catalysts.

Table 1
Detailed performance parameters of different CoS_2/MoS_2 heterostructure catalysts.

Catalyst	Preparation condition	Electrolyte	Current density (j, mA cm^{-2})	Overpotential (mV)	Tafel (mV dec^{-1})	R_{ct} (Ω)	C_{dl} (mF cm^{-2})	Ref.
CoS_2-MoS_2	Hydrothermal and anneal	1.0 M KOH	10	130	66.8	73.2	35.7	[42]
CoS_2/MoS_2	Hydrothermal	0.5 M H_2SO_4	10	154	61	305	77	[43]
CoS_2@MoS_2	Hydrothermal	0.50 M H_2SO_4	10	276	60		2.79	[44]
CoS_2/MoS_2	Reflux and anneal	1.0 M KOH	10	170	127			[45]
CoS_2-MoS_2	Hydrothermal, anneal and hydrothermal	1.0 M KOH	10	251	92		4.2	[46]
		0.5 M H_2SO_4		218	90		9.3	
CoS_2-MoS_2 MSHSs	Anneal, hydrothermal and anneal	1.0 M KOH	10	109	52			[47]
CoS_2-MoS_2 HNAs/Ti	Etching and anneal	1.0 M KOH	10	82	59			[41]
MoS_2/CoS_2 NTs	Chemical soaking, hydrothermal, anneal, hydrothermal and anneal	0.5 M H_2SO_4	10	90	30			[48]
		1.0 M KOH		85	34			
		1.0 M PBS		150	66.1			
CoS2/MoS_2-5	Hydrothermal	1.0 M KOH	10	73	80.42	1.199	456.84	This work

The Fermi energy level of CoS_2 crosses the energy band with a forbidden bandwidth of 0, indicating that CoS_2 exhibits metallicity, possesses strong electrical conductivity, and is an excellent candidate for constructing heterostructures with MoS_2. Indeed, enhanced catalytic activities have been observed in MoS_2/CoS_2 heterostructures through sophisticated nanotechnology [52]. Meanwhile, when observing the energy band structure of the CoS_2/MoS_2 heterojunction, the Fermi energy level passes through the energy band, and the nearby energy bands are relatively flat. This indicates that the system is metallic, has strong electronic localization, and is relatively stable. Fig. 7d–f illustrate the elemental projected density of states plots of MoS_2, CoS_2, and the CoS_2/MoS_2 heterojunction. By comparing the DOS of the three

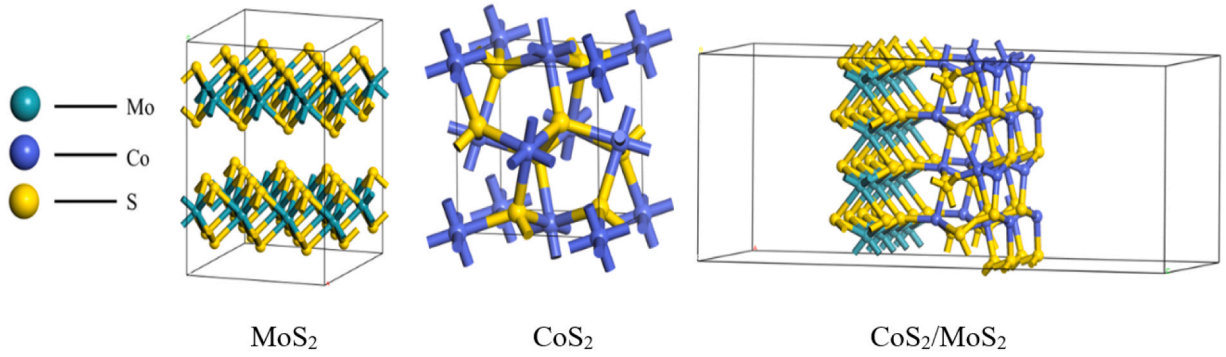


Fig. 6. Molecular model for DFT simulation of MoS_2, CoS_2 and CoS_2/MoS_2 heterojunctions.

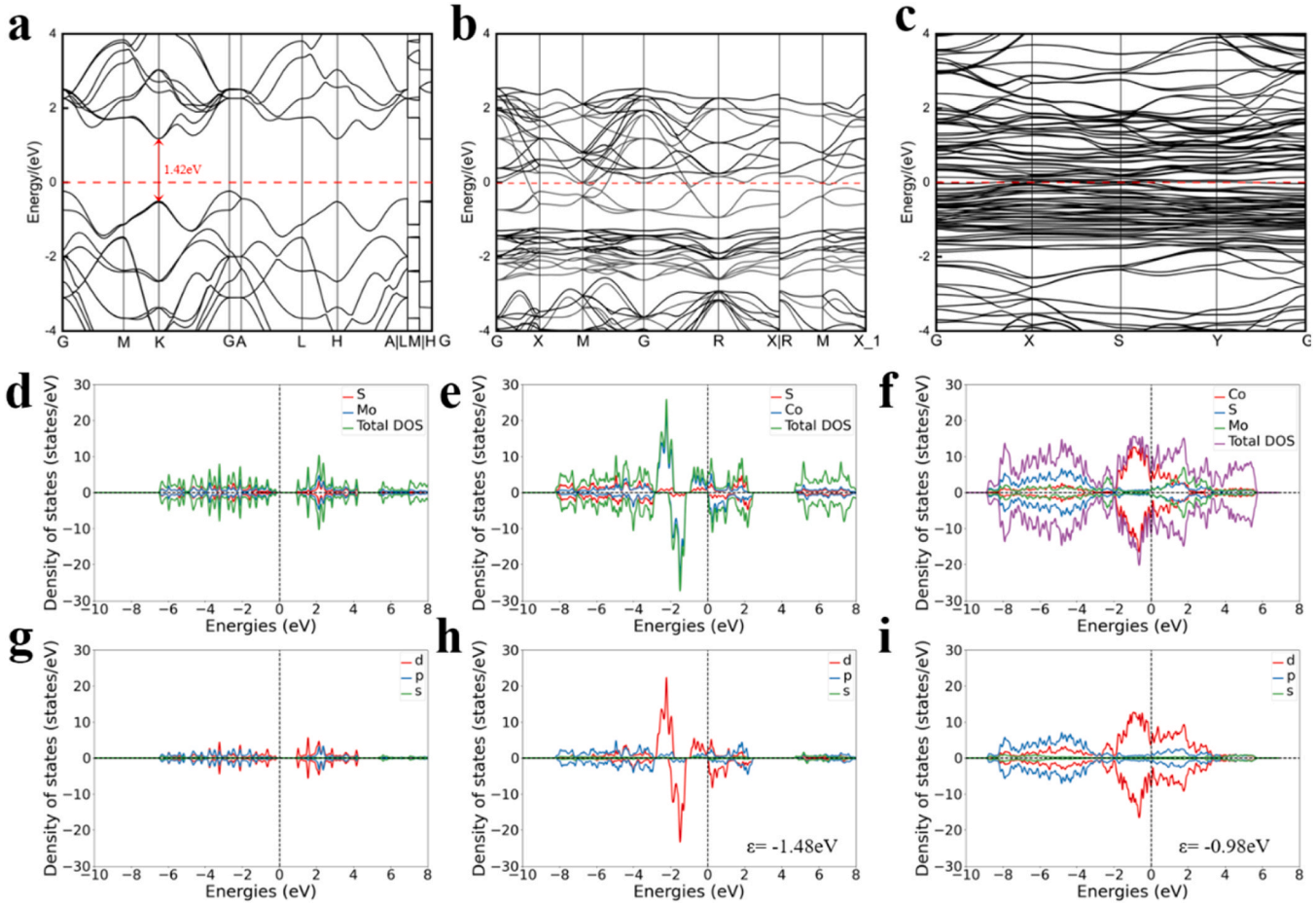


Fig. 7. (a–c) Band structures of MoS_2, CoS_2, and CoS_2/MoS_2 heterojunctions; (d–f) elemental projected density of states plots; (g–i) orbital projected density of states plots; ε:d-band center.

materials, it is evident that a new energy level emerges near the Fermi energy level following the formation of the CoS_2/MoS_2 heterojunction, with a more pronounced contribution from the Co atoms. Fig. 7g–i show the density of orbital projection states of MoS_2, CoS_2, and CoS_2/MoS_2 heterojunctions, with calculated d-band centers of − 1.48 eV for CoS_2 and − 0.98 eV for the CoS_2/MoS_2 heterojunction, respectively. The d-band center of the heterojunction is significantly shifted upward relative to CoS_2, which is closer to the Fermi energy level. According to the d-band center theory [53] proposed by Hammer and Nørskov, the higher the d-band of the transition metal is relative to the Fermi level, the higher the antibonding energy level is relative to the Fermi level. Consequently, the lower the degree of the antibonding energy level being occupied by the electrons, the weaker the bonding becomes. This results in stronger adsorption of H* in the HER and enhances the catalytic activity.

Meanwhile, in this paper, the free energies of hydrogen adsorption are shown in Fig. 8. MoS_2, CoS_2, and CoS_2/MoS_2 heterojunctions were calculated (Fig. 8b), modelled using the (200) face of CoS_2 and the (002) face of MoS_2 as the catalytic surfaces. Hydrogen adsorption on the CoS_2/MoS_2 heterojunctions was modelled at the cross-interface. ΔG for MoS_2 and CoS_2 is 1.52 eV and 0.44 eV, respectively. In contrast, the ΔG of the CoS_2/MoS_2 heterojunction is only − 0.08 eV, which is very close to 0. According to the Sabatier principle, this ensures that hydrogen adsorption and desorption on the surface of the CoS_2/MoS_2 catalyst are faster,

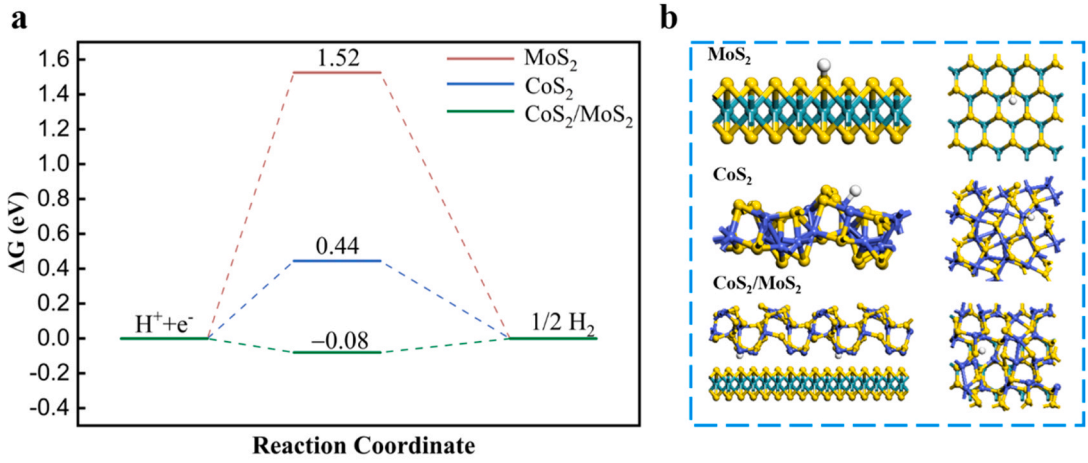


Fig. 8. (a) Free energy diagrams of hydrogen adsorption on MoS_2, CoS_2 and CoS_2/MoS_2 heterojunction. (b) Model of hydrogen adsorption on MoS_2, CoS_2, CoS_2/MoS_2 heterojunction.

enhancing the rate of HER.

4. Conclusion

In summary, the high-performance CoS_2/MoS_2 heterostructured nanosheets catalyst (CoS_2/MoS_2-5) was prepared on carbon cloth by modulating the electronic structure by continuously adjusting the molar ratio of ammonium molybdate to cobalt nitrate. The loose porous structure formed by CoS_2/MoS_2-5 heterostructured nanosheets showed optimized electronic interactions, providing abundant electrocatalytic active sites and fast diffusion pathways for reactants and intermediates. CoS_2/MoS_2-5 exhibited excellent HER performance in the alkaline electrolyte, with overpotentials of 73, 112, and 144 mV at 10, 20, and 50 mA cm^{-2}, respectively. Moreover, it demonstrated good stability during constant-current-density HER tests at 10 and 100 mA cm^{-2} for over 100 h. In addition, DFT theoretical calculations demonstrate that the redistribution of electrons occurring at the interface of the CoS_2/MoS_2 heterostructure improves the material's electrical conductivity. This enhancement results in a better Gibbs free energy for hydrogen adsorption, leading to faster adsorption and detachment of hydrogen on its surface, thereby greatly enhancing its HER rate. This work provides an experimental and theoretical basis for designing and developing efficient and highly active heterostructured HER electrocatalysts by modulating electronic structure.

CRediT authorship contribution statement

Jiajia Lian: Software, Formal analysis. **Baozhong Liu:** Supervision, Funding acquisition. **Wenyan Bi:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Yikai Hou:** Software. **Menglin Yu:** Software. **Tianen Li:** Resources. **Qiang Zhou:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Yihong Sun:** Software, Formal analysis. **Jianfeng Wan:** Writing – review & editing, Methodology, Conceptualization. **Shizheng Xie:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2024.175847.

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